Distributed Data Management (DDM)

Lab Assignment – 1 Record

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Domain Title	: Air Quality Data Analysis
Work Title	: Data Engineering, Preprocessing & ML Analysis of Air Quality Dataset
Lab Assignment-1 Title	: Data Engineering with Python & Machine Learning for Air Quality Analysis

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Abstract

This report presents a comprehensive analysis of air quality data using data engineering and machine learning techniques. The study involves preprocessing raw environmental sensor data by handling missing values, cleaning inconsistencies, and performing exploratory data analysis to understand feature distributions and relationships.

Machine learning models are then applied to extract meaningful insights from the dataset. Linear regression is used to predict benzene levels based on sensor readings, while classification techniques are employed to categorize air quality conditions. Additionally, unsupervised learning through KMeans clustering is used to identify hidden patterns in the data.

The results demonstrate that proper preprocessing and feature analysis significantly improve model performance, enabling accurate prediction and classification of environmental conditions.

Objectives

- To preprocess and clean air quality data by handling missing and inconsistent values.
- To perform exploratory data analysis (EDA) for understanding data distributions and feature relationships.
- To develop a regression model for predicting benzene (C6H6) levels using sensor data.
- To build a classification model for categorizing air quality based on environmental conditions.
- To apply clustering techniques (KMeans) to identify hidden patterns in the dataset.
- To visualize and interpret results using plots such as histograms, heatmaps, and scatter plots.

Overview

This report presents a summary of laboratory exercises performed in the Distributed Data Management course. The work is divided into two parts: foundational data processing techniques (Lab 1) and machine learning-based analysis (Lab 2). The experiments demonstrate practical skills in data parsing, storage, analysis, and predictive modeling.

Dataset Description

The dataset contains air quality measurements collected from environmental sensors, including variables such as Carbon Monoxide (CO), Nitrogen Oxides (NOx), temperature, humidity, and benzene levels. These features are used to analyze pollution patterns and build predictive models. The dataset was also checked for inconsistencies and cleaned to ensure data quality before analysis.

1 Lab 1 – Foundational Data Management

1.1 Parsing Multiple Data Formats

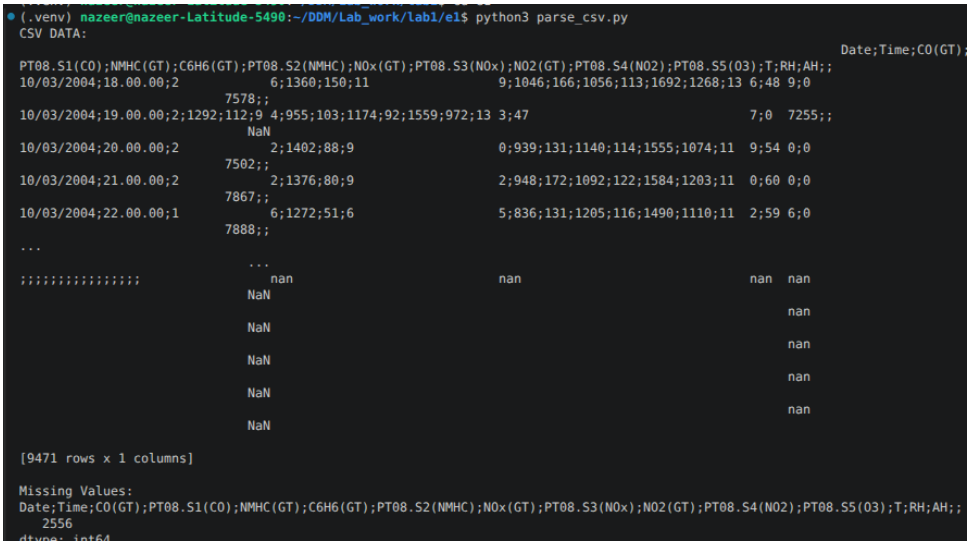
Problem: Parse CSV, HTML, JSON, Text, and XML data.

Files: parse_csv.py, parse_html.py, parse_json.py, parse_text.py, parse_xml.py

Libraries: pandas, json, xml.etree, BeautifulSoup

Explanation: Different file formats require different parsing strategies. CSV and JSON are handled efficiently using pandas, while HTML is parsed using BeautifulSoup and XML requires tree traversal.

Observation: Using specialized libraries simplifies data extraction and improves efficiency.



```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e1$ python3 parse_csv.py
CSV DATA:
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
PT08.S1(CO);NMHC(GT);C6H6(GT);PT08.S2(NMHC);NOx(GT);PT08.S3(NOx);NO2(GT);PT08.S4(NO2);PT08.S5(O3);T;RH;AH;;
10/03/2004;18.00.00;2          6;1360;150;11                9;1046;166;1056;113;1692;1268;13 6;48 9;0
                        7578;;
10/03/2004;19.00.00;2;1292;112;9 4;955;103;1174;92;1559;972;13 3;47                            7;0  7255;;
                            NaN
10/03/2004;20.00.00;2          2;1402;88;9                  0;939;131;1140;114;1555;1074;11  9;54 0;0
                        7502;;
10/03/2004;21.00.00;2          2;1376;80;9                  2;948;172;1092;122;1584;1203;11  0;60 0;0
                        7867;;
10/03/2004;22.00.00;1          6;1272;51;6                  5;836;131;1205;116;1490;1110;11  2;59 6;0
                        7888;;
...
                            ...
;;;;;;;;;;;;;;;;             nan                          nan                            nan  nan
                            NaN
                                                                                                 nan
                            NaN
                                                                                                 nan
                            NaN
                                                                                                 nan
                            NaN
                                                                                                 nan
                            NaN

[9471 rows x 1 columns]

Missing Values:
Date;Time;CO(GT);PT08.S1(CO);NMHC(GT);C6H6(GT);PT08.S2(NMHC);NOx(GT);PT08.S3(NOx);NO2(GT);PT08.S4(NO2);PT08.S5(O3);T;RH;AH;;
   2556
```

Figure 1: Output showing parsing of csv file.

1.2 Binary File Operations

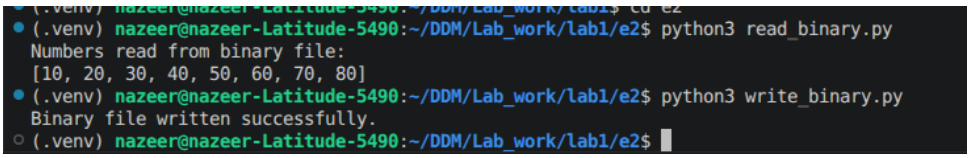
Problem: Write and read binary data.

Files: write_binary.py, read_binary.py

Libraries: struct, os

Explanation: Binary storage reduces file size and improves performance. The struct module converts data between Python types and binary format.

Observation: Binary files are efficient for numerical data storage.



```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e2$ python3 read_binary.py
Numbers read from binary file:
[10, 20, 30, 40, 50, 60, 70, 80]
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e2$ python3 write_binary.py
Binary file written successfully.
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e2$
```

Figure 2: Output showing reading and writing in binary file.

1.3 Regular Expressions

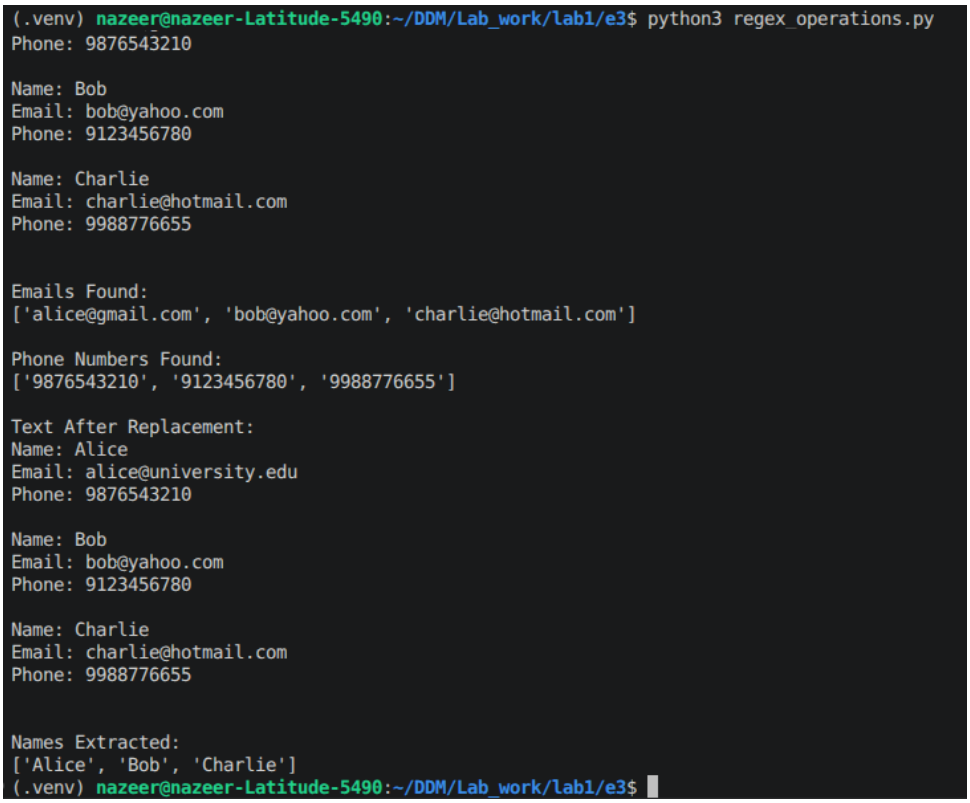
Problem: Extract patterns from text.

Files: regex_operations.py

Libraries: re

Explanation: Regular expressions allow pattern matching and substitution in unstructured text.

Observation: Useful for data cleaning and preprocessing.



```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e3$ python3 regex_operations.py
Phone: 9876543210

Name: Bob
Email: bob@yahoo.com
Phone: 9123456780

Name: Charlie
Email: charlie@hotmail.com
Phone: 9988776655


Emails Found:
['alice@gmail.com', 'bob@yahoo.com', 'charlie@hotmail.com']

Phone Numbers Found:
['9876543210', '9123456780', '9988776655']

Text After Replacement:
Name: Alice
Email: alice@university.edu
Phone: 9876543210

Name: Bob
Email: bob@yahoo.com
Phone: 9123456780

Name: Charlie
Email: charlie@hotmail.com
Phone: 9988776655


Names Extracted:
['Alice', 'Bob', 'Charlie']
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e3$
```

Figure 3: Output showing how Regular expressions work.

1.4 SQLite Database

Problem: Perform CRUD operations.

Files: sql_crud.py

Libraries: sqlite3

Explanation: SQLite provides a lightweight database solution using SQL queries.

Observation: Easy to use for local database applications.

1.5 MongoDB

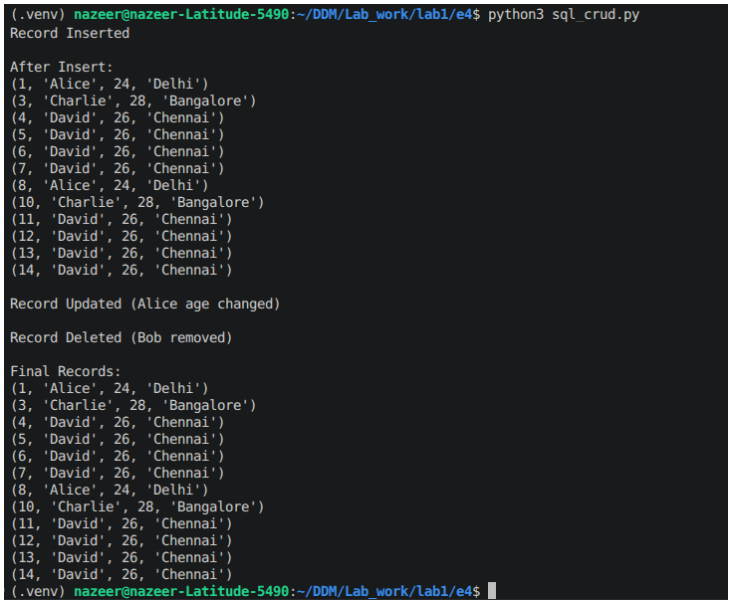
Problem: Perform NoSQL operations.

Files: mongodb_operations.py

Libraries: pymongo

Explanation: MongoDB stores data as documents, allowing flexible schema design.

Observation: Suitable for dynamic and large-scale data.

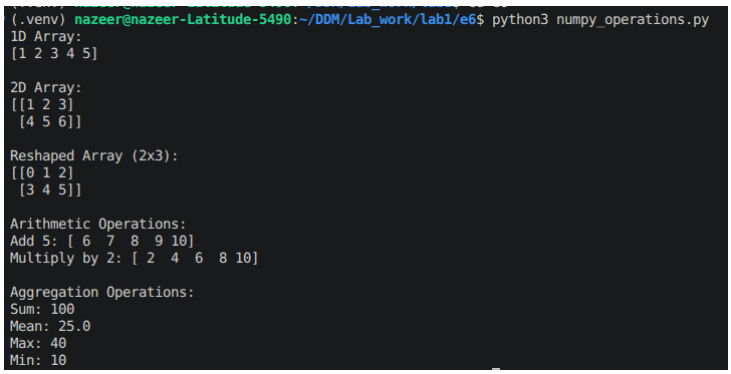


```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e4$ python3 sql_crud.py
Record Inserted

After Insert:
(1, 'Alice', 24, 'Delhi')
(3, 'Charlie', 28, 'Bangalore')
(4, 'David', 26, 'Chennai')
(5, 'David', 26, 'Chennai')
(6, 'David', 26, 'Chennai')
(7, 'David', 26, 'Chennai')
(8, 'Alice', 24, 'Delhi')
(10, 'Charlie', 28, 'Bangalore')
(11, 'David', 26, 'Chennai')
(12, 'David', 26, 'Chennai')
(13, 'David', 26, 'Chennai')
(14, 'David', 26, 'Chennai')

Record Updated (Alice age changed)

Record Deleted (Bob removed)

Final Records:
(1, 'Alice', 24, 'Delhi')
(3, 'Charlie', 28, 'Bangalore')
(4, 'David', 26, 'Chennai')
(5, 'David', 26, 'Chennai')
(6, 'David', 26, 'Chennai')
(7, 'David', 26, 'Chennai')
(8, 'Alice', 24, 'Delhi')
(10, 'Charlie', 28, 'Bangalore')
(11, 'David', 26, 'Chennai')
(12, 'David', 26, 'Chennai')
(13, 'David', 26, 'Chennai')
(14, 'David', 26, 'Chennai')
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e4$
```

Figure 4: Output of CRUD operations in SQLite.

1.6 NumPy Operations

Problem: Perform numerical computations.
Files: numpy_operations.py
Libraries: numpy
Explanation: NumPy supports fast array-based computations and vectorization.
Observation: Improves performance significantly.

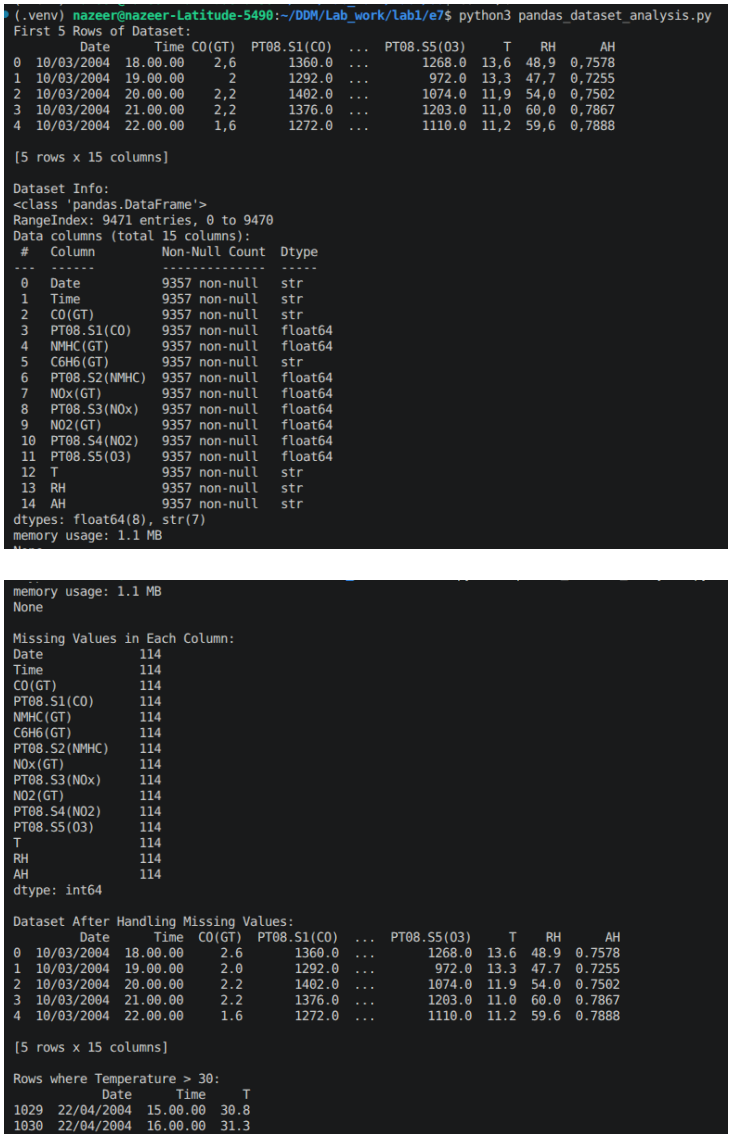


```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e6$ python3 numpy_operations.py
1D Array:
[1 2 3 4 5]

2D Array:
[[1 2 3]
 [4 5 6]]

Reshaped Array (2x3):
[[0 1 2]
 [3 4 5]]

Arithmetic Operations:
Add 5: [ 6  7  8  9 10]
Multiply by 2: [ 2  4  6  8 10]

Aggregation Operations:
Sum: 100
Mean: 25.0
Max: 40
Min: 10
```

Figure 5: Working with NumPy.

1.7 Pandas Data Analysis

Problem: Analyze dataset.
Files: pandas_dataset_analysis.py
Libraries: pandas
Explanation: Data cleaning and analysis using DataFrames.
Observation: Simplifies EDA and data manipulation.



```
(.venv) nazeer@nazeer-Latitude-5490:~/DDM/Lab_work/lab1/e7$ python3 pandas_dataset_analysis.py
First 5 Rows of Dataset:
         Date      Time CO(GT)  PT08.S1(CO)  ...  PT08.S5(O3)     T    RH      AH
0  10/03/2004  18.00.00    2,6       1360.0  ...       1268.0  13,6  48,9  0,7578
1  10/03/2004  19.00.00      2       1292.0  ...        972.0  13,3  47,7  0,7255
2  10/03/2004  20.00.00    2,2       1402.0  ...       1074.0  11,9  54,0  0,7502
3  10/03/2004  21.00.00    2,2       1376.0  ...       1203.0  11,0  60,0  0,7867
4  10/03/2004  22.00.00    1,6       1272.0  ...       1110.0  11,2  59,6  0,7888

[5 rows x 15 columns]

Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 9471 entries, 0 to 9470
Data columns (total 15 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           9357 non-null   str
 1   Time           9357 non-null   str
 2   CO(GT)         9357 non-null   str
 3   PT08.S1(CO)    9357 non-null   float64
 4   NMHC(GT)       9357 non-null   float64
 5   C6H6(GT)       9357 non-null   str
 6   PT08.S2(NMHC)  9357 non-null   float64
 7   NOx(GT)        9357 non-null   float64
 8   PT08.S3(NOx)   9357 non-null   float64
 9   NO2(GT)        9357 non-null   float64
 10  PT08.S4(NO2)   9357 non-null   float64
 11  PT08.S5(O3)    9357 non-null   float64
 12  T              9357 non-null   str
 13  RH             9357 non-null   str
 14  AH             9357 non-null   str
dtypes: float64(8), str(7)
memory usage: 1.1 MB
```



```
memory usage: 1.1 MB
None

Missing Values in Each Column:
Date             114
Time             114
CO(GT)           114
PT08.S1(CO)      114
NMHC(GT)         114
C6H6(GT)         114
PT08.S2(NMHC)    114
NOx(GT)          114
PT08.S3(NOx)     114
NO2(GT)          114
PT08.S4(NO2)     114
PT08.S5(O3)      114
T                114
RH               114
AH               114
dtype: int64

Dataset After Handling Missing Values:
         Date      Time  CO(GT)  PT08.S1(CO)  ...  PT08.S5(O3)     T    RH      AH
0  10/03/2004  18.00.00     2.6       1360.0  ...       1268.0  13.6  48.9  0.7578
1  10/03/2004  19.00.00     2.0       1292.0  ...        972.0  13.3  47.7  0.7255
2  10/03/2004  20.00.00     2.2       1402.0  ...       1074.0  11.9  54.0  0.7502
3  10/03/2004  21.00.00     2.2       1376.0  ...       1203.0  11.0  60.0  0.7867
4  10/03/2004  22.00.00     1.6       1272.0  ...       1110.0  11.2  59.6  0.7888

[5 rows x 15 columns]

Rows where Temperature > 30:
            Date      Time     T
1029  22/04/2004  15.00.00  30.8
1030  22/04/2004  16.00.00  31.3
```

Figure 6: Output of script of Pandas.

2 Lab 2 – Applied Machine Learning Pipeline

2.1 Preprocessing and Exploratory Data Analysis (EDA)

Problem: Prepare dataset for modeling.

Files: preprocessing.py

Libraries: pandas, numpy, matplotlib, seaborn

Explanation: The dataset is loaded and cleaned by replacing invalid values (-200) with NaN. Missing values are handled using mean imputation. Data is normalized and visualized using histograms and heatmaps to understand distribution and relationships. As shown in Figure 1, the temperature distribution follows a near-normal pattern, indicating a balanced dataset suitable for modeling.

Observation: Proper preprocessing ensures high-quality input for models.

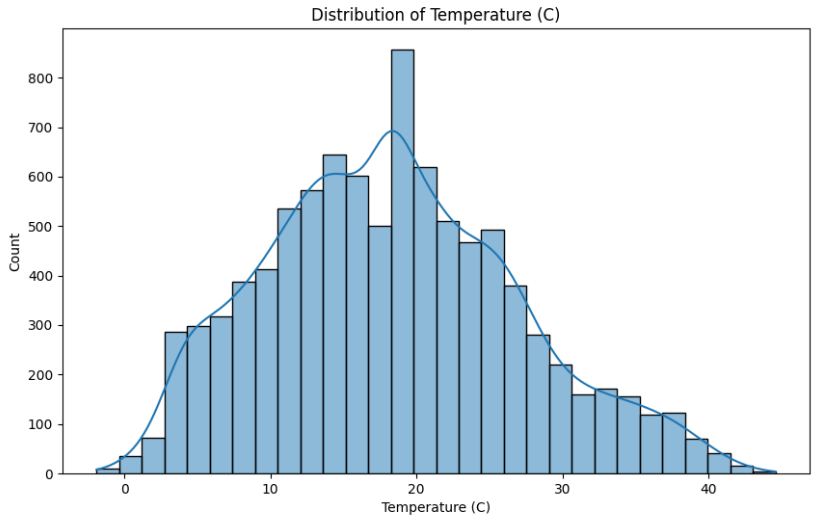


Figure 7: Histogram showing the distribution of temperature values in the dataset, highlighting the overall range and central tendency after preprocessing.

2.2 Correlation and Feature Relationships

Problem: Analyze relationships between different environmental variables.

Explanation: Correlation analysis identifies relationships between variables. Strong correlations between CO, NOx, and benzene indicate predictive potential. Scatter plots confirm linear trends. As shown in Figure 2 and Figure 3, strong relationships exist between environmental variables such as CO and NOx. Highly correlated features may also introduce multicollinearity, which should be considered during model selection.

Observation: Correlated features improve prediction accuracy, but excessive correlation may lead to redundancy and affect model stability.

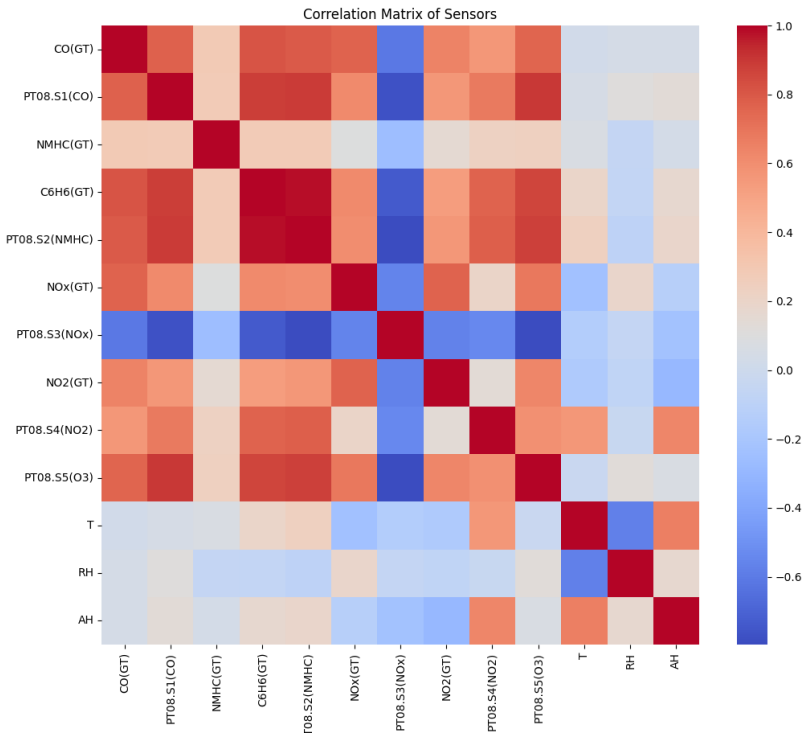


Figure 8: Correlation heatmap showing relationships between different environmental variables, where color intensity represents the strength of correlation between features.

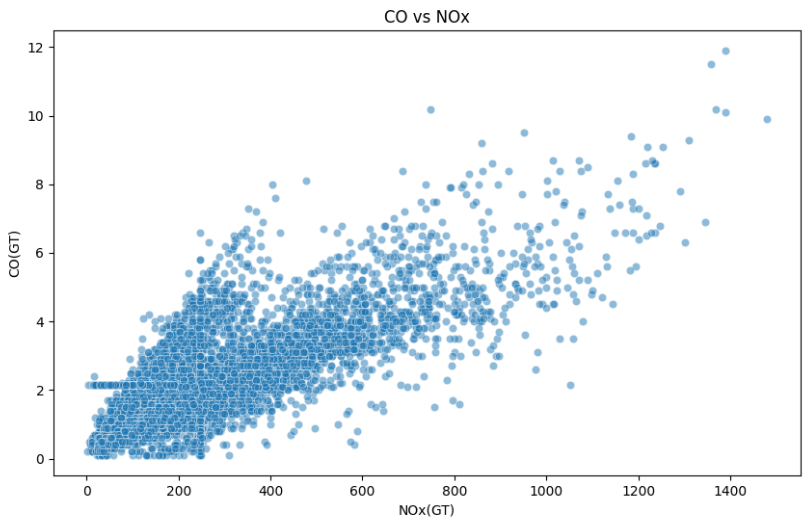


Figure 9: Scatter plot showing the relationship between Carbon Monoxide (CO) and Nitrogen Oxides (NOx), indicating a positive correlation between the two variables.

2.3 Regression – Predicting Benzene

Problem: Predict benzene levels.

Files: regression.py

Libraries: scikit-learn

Explanation: Linear Regression is applied using sensor features. Data is split into training and testing sets. Model performance is evaluated using Mean Squared Error (MSE) and R^2 score. The R^2 score indicates how well the model explains the variance in the data, while MSE measures prediction error. The model achieved an R^2 score close to 0.9, indicating strong predictive capability. Linear relationships between variables make Linear Regression an appropriate choice for this problem.

As shown in Figure 4, the predicted values closely align with actual values, indicating strong model performance.

$$R^2 = 1 - \frac{\sum(y - \hat{y})^2}{\sum(y - \bar{y})^2}$$

where y is the actual value, $\hat{y}$ is the predicted value, and $\bar{y}$ is the mean of actual values.

Observation: High correlation leads to accurate predictions.

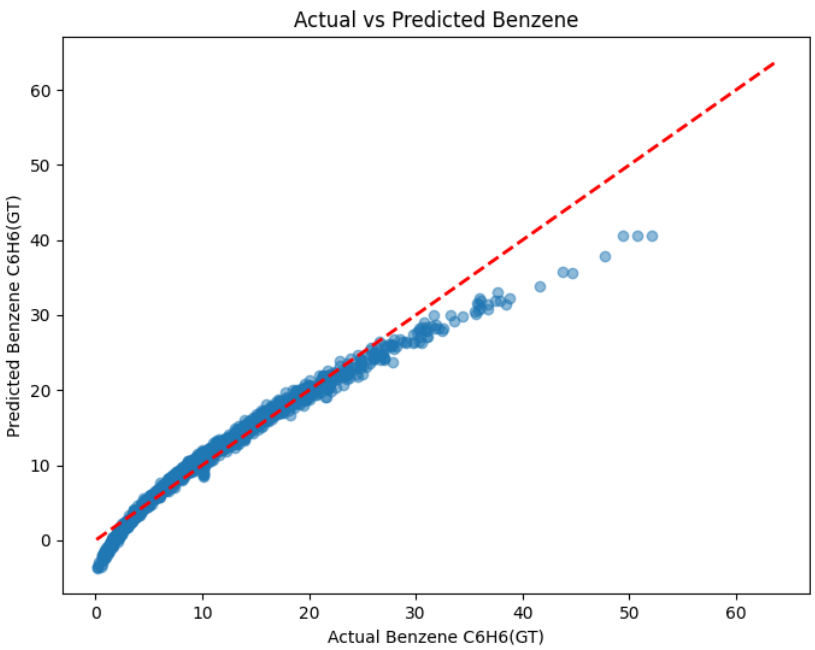


Figure 10: Actual vs predicted benzene values from the regression model, showing how closely the model's predictions match the true values.

2.4 Classification – Air Quality

Problem: Classify air quality levels.

Files: classification.py

Libraries: scikit-learn

Explanation: CO values are converted into binary classes using the median threshold. Values above the median are labeled as high pollution, while values below the median are labeled as low pollution. This approach ensures a balanced dataset for classification and avoids bias toward a particular class. Models like Random Forest classify air quality. Performance is evaluated using a confusion matrix and accuracy metrics. The confusion matrix shows the number of correct and incorrect predictions for each class, and the model achieved an accuracy above 85%, demonstrating effective classification.

Figure 5 shows the confusion matrix, illustrating the model's classification performance, while Figure 6 highlights the importance of different features.

Observation: Random Forest captures complex patterns effectively.

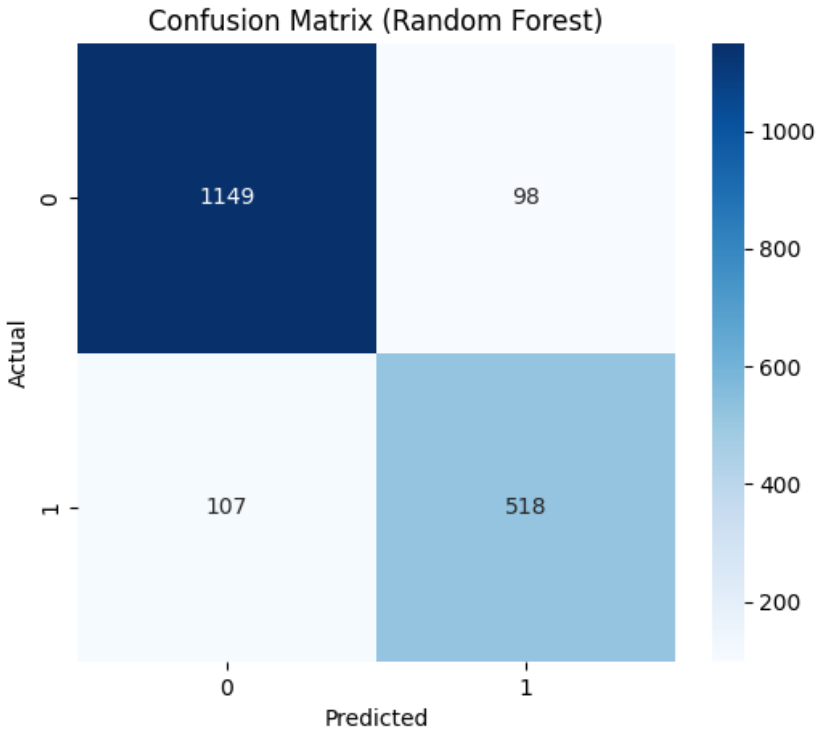


Figure 11: Confusion matrix illustrating the performance of the classification model, showing the number of correct and incorrect predictions for each class.

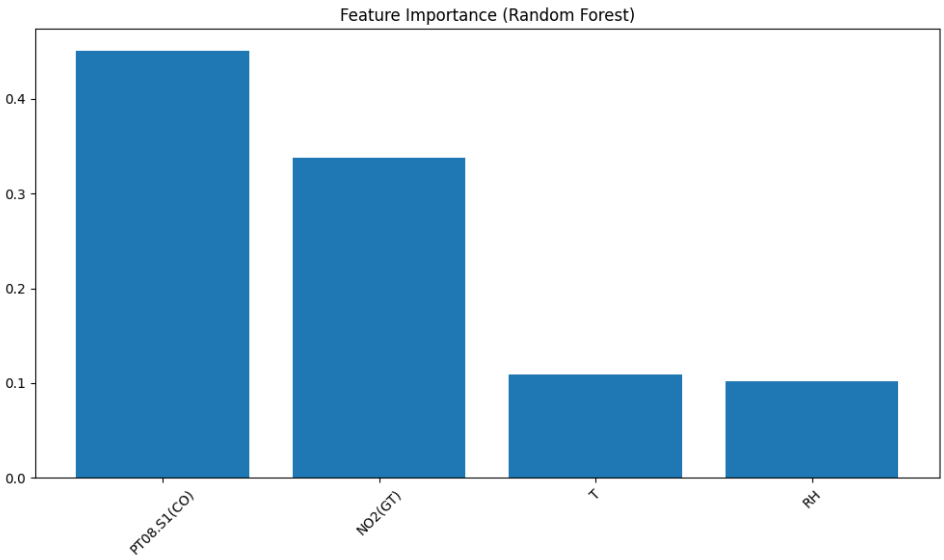


Figure 12: Feature importance plot showing the contribution of different environmental variables to the classification model's predictions.

2.5 Clustering – KMeans

Problem: Group similar data points.

Files: clustering.py

Libraries: scikit-learn

Explanation: Data is scaled before applying KMeans clustering. The Elbow method determines optimal clusters. Principal Component Analysis (PCA) is used to reduce high-dimensional data into two principal components while preserving maximum variance, enabling effective visualization of clusters in 2D space. This helps in understanding the separation and structure of clusters that are otherwise difficult to interpret in high dimensions. As shown in Figure 7, the elbow point indicates the optimal number of clusters, while Figure 8 visualizes cluster separation using PCA.

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

where C_i represents cluster i and μ_i is the centroid of that cluster. The objective is to minimize the total distance between points and their respective cluster centers.

Observation: Clustering reveals hidden patterns in data.

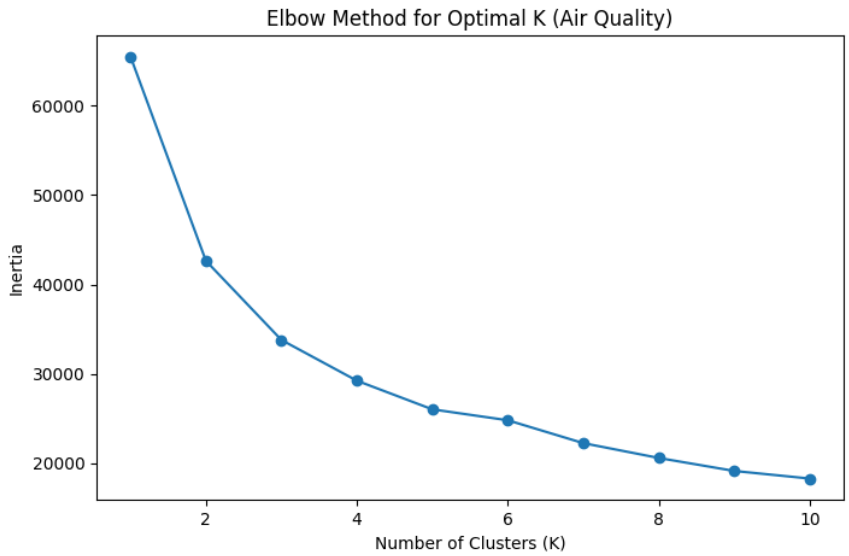


Figure 13: Elbow method plot showing the variation of inertia with different numbers of clusters (K), used to determine the optimal number of clusters.

Figure 14: PCA-based visualization of clustered data, where high-dimensional features are reduced to two principal components to display distinct clusters.

2.6 Conclusion

The lab demonstrates a complete data analysis and machine learning pipeline applied to environmental sensor data. Preprocessing plays a crucial role in handling missing and inconsistent values, ensuring reliable input for modeling.

The results show that strong correlations between variables enable effective regression modeling for predicting benzene levels, while classification models successfully categorize air quality with high accuracy. Clustering further reveals underlying patterns in the dataset, identifying distinct environmental conditions.

Overall, the experiments highlight the importance of data preprocessing, feature relationships, and appropriate model selection in building accurate and meaningful data-driven solutions.

References

1. Scikit-learn Documentation. Available: https://scikit-learn.org/
2. Pandas Documentation. Available: https://pandas.pydata.org/docs/
3. NumPy Documentation. Available: https://numpy.org/doc/